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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 58.5

$$\begin{aligned}
 & \int \frac{dx}{3 + \cos x} \\
 & t = \tan \frac{x}{2} \\
 & \left\{ \begin{aligned} dt &= \frac{1}{2} \sec^2 \frac{x}{2} dx \\ dx &= 2 \cos^2 \frac{x}{2} dt = \frac{2}{1 + \tan^2 \frac{x}{2}} dt = \frac{2}{1 + t^2} dt \\ \cos x &= \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{1 - t^2}{1 + t^2} \end{aligned} \right. \\
 & \int \frac{dx}{3 + \cos x} = \int \frac{\frac{2}{1 + t^2} dt}{3 + \frac{1 - t^2}{1 + t^2}} \\
 & = \int \frac{\frac{2}{1 + t^2}}{\frac{3(1 + t^2) + 1 - t^2}{1 + t^2}} dt \\
 & = \int \frac{2}{3(1 + t^2) + 1 - t^2} dt \\
 & = \int \frac{2}{4 + 2t^2} dt \\
 & = \int \frac{1}{2 + t^2} dt \\
 & = \frac{1}{\sqrt{2}} \operatorname{Bgtan} \frac{t}{\sqrt{2}} + C \\
 & \tan \frac{x}{2} = t \\
 & \int \frac{dx}{3 + \cos x} = \frac{1}{\sqrt{2}} \operatorname{Bgtan} \left(\frac{\tan \frac{x}{2}}{\sqrt{2}} \right) + C
 \end{aligned}$$

References